Project: Worked examples
Title: Shaft: static, fatigue, keys, critical speed
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1. Scope and method

A stepped (optionally hollow) shaft on two bearings carrying radial loads in two planes and a steady torque: bending moment, slope and deflection along it, the fatigue and yield safety factors at the sections the user marks, slopes at the bearings, deflection at the loads, the first lateral critical speed and parallel keys.

WHAT IT CALCULATES	RELATION
Shaft bending moment	the shaft as an S2 beam (pin at A, roller at B) in each plane; $M = \sqrt{(M_y^2 + M_z^2)}$
Stress at a section	$\sigma_a = K_f \cdot 32M_a \cdot d/(\pi(d^4 - d_i^4))$, $\tau_m = K_{fs} \cdot 16T_m \cdot d/(\pi(d^4 - d_i^4))$; $K_f = 1 + q(K_t - 1)$
Endurance limit	$S_e = k_a \cdot k_b \cdot k_c \cdot k_d \cdot k_e \cdot k_f \cdot S_{e1}$, S_{e1} (the rotating-beam limit S_e') = $0.5 \cdot S_{ut}$ ($S_{ut} \le 1400$ MPa, else 700 MPa); $k_a = a \cdot S_{ut}$^b, $k_b = 1.24d^{-0.107}$ or $1.51d^{-0.157}$, $k_e = 1 - 0.08 \cdot z_R$
Fatigue (DE-Goodman)	$1/n_f = \sigma_a/S_e + \sqrt{3} \cdot \tau_m/S_{ut}$; yield $n_y = S_y/\sqrt{((\sigma_a)^2 + 3\tau_{pk}^2)}$, τ_{pk} from the larger of T and the peak torque T_{pk} when given
Critical speed	Rayleigh $\omega = \sqrt{(g \cdot \Sigma m \cdot y/\Sigma m \cdot y^2)}$ (upper bound); Dunkerley $1/\omega^2 = \Sigma\, m_i \cdot \delta_{ii}$ (lower bound, used for the ratio to the operating speed)
Parallel key	$\tau = 2T/(d \cdot w \cdot L)$, $\sigma = 4T/(d \cdot h \cdot L)$; required length $L = \max(2T \cdot n_{target}/(d \cdot w \cdot 0.577 \cdot S_y), 4T \cdot n_{target}/(d \cdot h \cdot S_y))$

Basis: Fatigue, shaft stresses, deflection, critical speeds, keys[1](10th ed., Ch. 6 (§6-9, Table 6-2 Marin surface factors), Ch. 7 (§7-4 to §7-7)); Shaft design[2](6th ed., Ch. 12).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
segments	Shaft segments (left to right)	5 row(s)		drawing: GBX-SH-01 rev C	any
x_A	Bearing A position	20	mm	drawing: GBX-SH-01 rev C	0 … 3000 mm
x_B	Bearing B position	280	mm	drawing: GBX-SH-01 rev C	1 … 3000 mm
loads	Radial loads and masses	1 row(s)		spec: Gear mesh forces from the gear calculation	any
T	Torque transmitted	0.3	kN·m	spec: Motor rated torque × service factor (E1)	0 … 50 kN·m

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
x_{T1}	Torque enters at	150	mm	drawing: GBX-SH-01 rev C	0 … 3000 mm
x_{T2}	Torque leaves at	330	mm	drawing: GBX-SH-01 rev C	0 … 3000 mm
n	Operating speed	1450	rpm	spec: Motor nameplate	0 … 30000 rpm
sections	Sections to check	4 row(s)		assumption: Shoulders r/d ≈ 0.05; keyseat at the pinion (K_t 2.14, K_ts 3.0); q from the Neuber fit.	any
S_{ut}	Ultimate tensile strength	700	MPa	spec: EN 10083-3 42CrMo4 +QT, certificate	300 … 2000 MPa
S_y	Yield strength	500	MPa	spec: EN 10083-3 42CrMo4 +QT, certificate	150 … 1900 MPa
E	Young's modulus	207	GPa	assumption: Steel.	50 … 250 GPa
ρ	Density (shaft self-mass in the critical speed)	7850	kg/m³	assumption: Steel.	0 … 20000 kg/m³
finish	Surface finish	machined or cold-drawn		drawing: GBX-SH-01 rev C	any
R	Reliability	99	%	assumption: Design reliability.	50 … 99.9 %
k_d	Temperature factor k_d	1		assumption: Below 250 °C.	0.5 … 1.1
k_f	Miscellaneous factor k_f	1		assumption: No other effects.	0.3 … 1.5
n_{target}	Target safety factor	1.5		spec: Company design standard	1.2 … 4
θ_{max}	Slope limit at the bearings	0.001	rad	assumption: Deep-groove ball bearing slope limit.	0.00001 … 0.1 rad
y_{max}	Deflection limit at the loads	0.13	mm	assumption: Spur gear deflection limit.	0.001 … 10 mm
x_{key}	Key position	150	mm	drawing: GBX-SH-01 rev C	0 … 3000 mm
w	Key width	14	mm	drawing: GBX-SH-01 rev C	1 … 100 mm
h	Key height	9	mm	drawing: GBX-SH-01 rev C	1 … 100 mm

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
L_{key}	Key length	40	mm	drawing: GBX-SH-01 rev C	1 … 500 mm
$S_{y,key}$	Key yield strength	400	MPa	spec: C45 key steel	150 … 1500 MPa

TABLE ROW	VALUES
segments[1]	x_end = 40 mm; d = 35 mm; d_i = 0 mm
segments[2]	x_end = 120 mm; d = 40 mm; d_i = 0 mm
segments[3]	x_end = 180 mm; d = 45 mm; d_i = 0 mm
segments[4]	x_end = 260 mm; d = 40 mm; d_i = 0 mm
segments[5]	x_end = 340 mm; d = 35 mm; d_i = 0 mm
loads[1]	x = 150 mm; F_y = -2.2 kN; F_z = 6 kN; m = 4 kg
sections[1]	x = 120 mm; K_t = 1.7; K_ts = 1.5; q = 0.8; q_s = 0.9
sections[2]	x = 150 mm; K_t = 2.14; K_ts = 3; q = 0.8; q_s = 0.9
sections[3]	x = 180 mm; K_t = 1.7; K_ts = 1.5; q = 0.8; q_s = 0.9
sections[4]	x = 260 mm; K_t = 1.6; K_ts = 1.4; q = 0.8; q_s = 0.9

3. Assumptions

A1. Linear elastic, two simple bearing supports (pin at A, roller at B); bending fully reversed by rotation, torque steady. (Shigley 10th ed. §7-4.)
A2. No torsional vibration; only the first lateral mode, by Rayleigh with Dunkerley as a lower bound. (Shigley 10th ed. §7-6.)
A3. K_t and K_{ts} come from user records or published fit equations; no stress-concentration chart is reproduced. (No stress-concentration chart is shipped; the factors are entered or derived.)
A4. At a shoulder the smaller diameter governs; k_c = 1 because the stresses are combined by von Mises. (Shigley 10th ed. §6-9, §7-4.)
A5. Loads, sections and steps closer together than 1/200 of the shaft length share one node. (Keeps the beam model well conditioned.)
A6. sections (Sections to check) is taken as 4 row(s). (Shoulders r/d ≈ 0.05; keyseat at the pinion (K_t 2.14, K_{ts} 3.0); q from the Neuber fit.)
A7. E (Young's modulus) is taken as 207 GPa. (Steel.)
A8. ρ (Density (shaft self-mass in the critical speed)) is taken as 7850 kg/m^3. (Steel.)
A9. R (Reliability) is taken as 99 %. (Design reliability.)
A10. k_d (Temperature factor k_d) is taken as 1. (Below 250 °C.)
A11. k_f (Miscellaneous factor k_f) is taken as 1. (No other effects.)
A12. θ_{max} (Slope limit at the bearings) is taken as 0.001 rad. (Deep-groove ball bearing slope limit.)
A13. y_{max} (Deflection limit at the loads) is taken as 0.13 mm. (Spur gear deflection limit.)

4. Working

[17] $R_A = \sqrt{(R_{A,y}^2 + R_{A,z}^2)}$
$= \sqrt{(1.1^2 + -3^2)}$
$= 3.1953$ kN
Radial reaction at bearing A
[18] $R_B = \sqrt{(R_{B,y}^2 + R_{B,z}^2)}$
$= \sqrt{(1.1^2 + -3^2)}$
$= 3.1953$ kN

Radial reaction at bearing B

[19] d_A = seat diameter at A

= x = 20 mm

= 35 mm

Shaft diameter at bearing A

[20] d_B = seat diameter at B

= x = 280 mm

= 35 mm

Shaft diameter at bearing B

[21] M_{max} = max $\sqrt{(M_y^2 + M_z^2)}$ along the shaft

= 37 stations

= 0.41539 kN·m

Largest bending moment

[22] θ_A = slope at A (two planes)

= x = 20 mm

= 0.0008961 rad

Slope at bearing A

[23] θ_B = slope at B (two planes)

= x = 280 mm

= 0.0008961 rad

Slope at bearing B

[24] y_{load} = largest deflection at a load

= 0.071775 mm at 150

= 0.071775 mm

Deflection at the loads

[26] $k_a = a \cdot S_{ut}\text{^}b$ [1](10th ed. §6-9, Table 6-2)

= 4.51 × 700^-0.265 (machined or cold-drawn)

= 0.79474

Surface factor

[27] $k_e = 1 - 0.08 \cdot z_R$ [1](10th ed. §6-9)

= 1 - 0.08 × 2.3263

= 0.81389

Reliability factor

[28] $S_{e1} = 0.5 \cdot S_{ut}$ (S_e')

= 700 MPa

= 350 MPa

Rotating-beam endurance limit

[29] X1 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T/(\pi d^3)$; $1/n_f = \sigma_a/S_e + \sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)

= x = 120 mm, d = 40 mm, k_b = 0.83561, S_e = 189.17 MPa, K_f = 1.56, K_{fs} = 1.45, M = 319.53 N·m, N·m, σ_a = 79.334 MPa, τ_m = 0 MPa, n_y = 6.3025

= 2.3845

Section at 120 mm: fatigue safety factor (DE-Goodman)

[30] X2 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T/(\pi d^3)$; $1/n_f = \sigma_a/S_e + \sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)

= x = 150 mm, d = 45 mm, k_b = 0.82514, S_e = 186.8 MPa, K_f = 1.912, K_{fs} = 2.8, M = 415.39 N·m, 300 N·m, σ_a = 88.778 MPa, τ_m = 46.947 MPa, n_y = 4.1532

= 1.6909

Section at 150 mm: fatigue safety factor (DE-Goodman)

[31] X3 = $\sigma_a = K_f \cdot 32M/(\pi d^3)$, $\tau_m = K_{fs} \cdot 16T/(\pi d^3)$; $1/n_f = \sigma_a/S_e + \sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)

= x = 180 mm, d = 40 mm, k_b = 0.83561, S_e = 189.17 MPa, K_f = 1.56, K_{fs} = 1.45, M = 319.53 N·m, 300 N·m, σ_a = 79.334 MPa, τ_m = 34.616 MPa, n_y = 5.0281

= 1.9801

Section at 180 mm: fatigue safety factor (DE-Goodman)

[32] X4 = $\sigma_a = K_f \cdot 32M/(\pi d^{3\prime})$, $\tau_m = K_{fs} \cdot 16T/(\pi d^{3\prime})$; $1/n_f = \sigma_a/S_e + \sqrt{3}\tau_m/S_{ut}$ [1](10th ed. §7-4)
= x = 260 mm, d = 35 mm, k_b = 0.84763, S_e = 191.9 MPa, K_f = 1.48, K_{fs} = 1.36, M = 63.906 N·m, 300 N·m, σ_a = 22.47 MPa, τ_m = 48.465 MPa, n_y = 5.7538
= 4.2192
Section at 260 mm: fatigue safety factor (DE-Goodman)

[33] $S_e = k_a \cdot k_b \cdot k_c \cdot k_d \cdot k_e \cdot k_f \cdot S_e'$ (governing section)
= k_c = 1
= 186.8 MPa
Endurance limit at the governing section

[34] $n_{f,min}$ = min over sections
= 4 section(s)
= 1.6909
Lowest fatigue safety factor

[35] $n_{y,min}$ = min over sections S_y/σ'_max
= 4 section(s)
= 4.1532
Lowest yield safety factor

[36] $n_c = \sqrt{(g \cdot \Sigma m \cdot y / \Sigma m \cdot y^2)}$ (Rayleigh) [1](10th ed. §7-6)
= Σm = 7.2337 kg over 37 nodes
= 39515 rpm
First lateral critical speed (Rayleigh)

[37] $n_{c,D} = 1/\omega^2 = \Sigma\, m_i \cdot \delta_{ii}$ (Dunkerley, lower bound) [1](10th ed. §7-6)
= 37 masses
= 37295 rpm
First critical speed, Dunkerley lower bound

[38] $r_c = n_{c,D}/n$
= 37295 / 1450
= 25.721
Critical-speed ratio (Dunkerley lower bound over operating speed)

[43] $\tau_{key} = 2T/(d \cdot w \cdot L)$ [1](10th ed. §7-7)
= 2 × 300 N·m / (45 × 14 × 40) mm³
= 23.81 MPa
Key shear stress

[44] $\sigma_{key} = 4T/(d \cdot h \cdot L)$ [1](10th ed. §7-7)
= 4 × 300 N·m / (45 × 9 × 40) mm³
= 74.074 MPa
Key bearing stress

[46] $L_{key,req} = \max(2T \cdot n_{target}/(d \cdot w \cdot 0.577 S_y),\ 4T \cdot n_{target}/(d \cdot h \cdot S_y))$
= n_{target} = 1.5
= 11.111 mm
Key length required

[47] $u_{key} = L_{req}/L$
= 11.111 / 40
= 0.27778
Key utilisation

[48] $u_f = n_{target}/n_{f,min}$
= 1.5 / 1.6909
= 0.88712
Fatigue utilisation

[49] $u_y = n_{target}/n_{y,min}$
= 1.5 / 4.1532
= 0.36117
Yield utilisation

[50] $u_{slope} = \max(\theta_A, \theta_B)/\theta_{max}$
$= 0.0008961 / 0.001$
$= 0.8961$
Slope utilisation
[51] $u_{defl} = y_{load}/y_{max}$
$= 0.071775 / 0.13\ mm$
$= 0.55212$
Deflection utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
R_A	3.1953 kN	Reaction at A
R_B	3.1953 kN	Reaction at B
d_A	35 mm	Seat diameter at A
d_B	35 mm	Seat diameter at B
M_{max}	0.41539 kN·m	Largest bending moment
θ_A	0.0008961 rad	Slope at A
θ_B	0.0008961 rad	Slope at B
y_{load}	0.071775 mm	Deflection at the loads
S_e	186.8 MPa	Endurance limit
$n_{f,min}$	1.6909	Lowest fatigue safety factor
$n_{y,min}$	4.1532	Lowest yield safety factor
n_c	39515 rpm	First critical speed (Rayleigh)
$n_{c,D}$	37295 rpm	Dunkerley lower bound
r_c	25.721	Critical-speed ratio
τ_{key}	23.81 MPa	Key shear
σ_{key}	74.074 MPa	Key bearing
$L_{key,req}$	11.111 mm	Key length required
u_{key}	0.27778	Key utilisation

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Slope at the bearings against limit	$u_{slope} = 0.8961$	$u_{slope} \le 1$	0.896	
PASS	Fatigue safety factor against target (DE-Goodman)	$u_f = 0.88712$	$u_f \le 1$	0.887	[1](10th ed. §7-4)
PASS	Deflection at the loads against limit	$u_{defl} = 0.55212$	$u_{defl} \le 1$	0.552	

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Yield safety factor against target	u_y = 0.36117	$u_y \le 1$	0.361	
PASS	Key length against required	u_{key} = 0.27778	$u_{key} \le 1$	0.278	

7. Applicability

WARNING — record: Tabulated values not taken from a reference record: S_{ut}, S_y, E (Shaft material); θ_{max}, y_{max} (Slope and deflection limits). Enter each as one so its source, edition and licence travel with it.

8. Verification

Template mech.shaft@1.1.1 Content hash 47130117080eab34 Engine calc-engine/1.0.0
Known-answer cases passed 3/3 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.
2. Mott, Machine Elements in Mechanical Design.